

MØDE

MATH + CODE

The autonomous company that teaches kids math by teaching them to code.



Kids are losing math — fast.

-15

POINTS

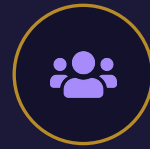
OECD average math score, 2018 → 2022 —
the steepest drop ever recorded, and
roughly three-quarters of a school year of
lost learning.

Source: OECD PISA 2022 (Vol. I)



Engagement, not ability

Most kids don't lack the capacity for math, they lose interest before mastery. Rote drills make it worse.



Teachers can't go 1:1

The thing that actually works is individual, adaptive attention and is exactly what a 30-child classroom can't deliver.



The gap compounds

Early disengagement snowballs into avoidance of every quantitative field downstream — STEM, finance, and AI itself.

Code is the best medium for math.

Writing code forces the abstract to become concrete: a variable is a number you can see change, a loop is a pattern you can run, geometry is something you draw. The evidence backs it — and so does the evidence for how we teach it.



$g \approx 0.49$

Transfer from coding to math

Meta-analysis of 105 studies: learning to program produces a moderate transfer benefit to mathematical and creative-reasoning skill.

Scherer et al., 2019



$g \approx 0.42$

Adaptive 1:1 tutoring vs. class

Meta-analysis of 14,321 learners: intelligent, one-to-one tutoring beats whole-class teaching — and rivals a human tutor.

Ma et al., 2014



Socratic

Never hand over the answer

MØDE's tutor escalates Question → Hint → Partial example → Explanation, keeping the child in the zone of proximal development.

MØDE pedagogy

One platform, three ages of builder.



Ages 5–7

Early Explorers

Block-based puzzles for pattern, spatial logic and number sense. Math before reading fluency.



Ages 8–12

Middle Builders

Python & JavaScript projects — build games and apps that require variables, geometry and algebra to work.



Ages 13–15

Teen Engineers

Text-based coding into early calculus, data analysis and the foundations of machine learning.



Embedded code editor

Solve in-browser, run instantly



Gamified progression

Points, badges, adventures



Parent dashboard

Live progress, auto-generated



Adaptive curriculum

Re-planned every session

Four real layers run the business.

The Hermes brief asks for an agent that earns, spends and operates. MØDE is built as exactly that — a company whose core functions are agent-driven, not just an app with a chatbot.



Hermes

THE BRAIN

Socratic tutor + business operator. Plans curriculum, talks to students, and makes the operating decisions.



Nemotron 3 Ultra

THE HORSEPOWER

NVIDIA model layer running inference intelligently and at the scale student demand requires.



NemoClaw

THE GUARDRAIL

NVIDIA safety layer that screens every generated lesson and message before a child ever sees it.



Stripe Skills

THE WALLET

Lets the agent charge subscriptions, pay its own bills, and provision the SaaS it needs to run.

A business loop that closes itself.



EARN



Parent subscribes → Stripe runs the charge



Failed payments retried & dunned automatically



Progress reports sent with invoice receipts



SPEND



Demand spikes → agent provisions more compute



Pays server & SaaS bills from a scoped Stripe key



Every outflow inside a hard spend cap

More students → more revenue → more compute the agent buys itself → capacity for more students.

Autonomous — with humans on the loop.



Lesson from Project Vend

When Anthropic let an agent run a shop unbounded, it dropped prices to zero and bought at a loss. Given spend caps, a CRM and structure, it improved sharply.

**MØDE's stance: unbounded autonomy fails.
Bounded autonomy — caps + approval points —
is what actually ships.**

Source: Anthropic Project Vend (Phase 1 & 2)



Spend caps + approvals

Hard ceilings on every outflow; large or unusual actions wait for a human.



Safety-screened output

NemoClaw checks every lesson and tutor message before it reaches a child.



Data minimization

Store pedagogical state — mastery, not raw chat transcripts.



Compliance by design

COPPA · UK Children's Code · GDPR · POPIA · FERPA

One slice, end to end, on camera.

01



A child codes

A learner draws a square in the editor — but the angles are wrong.

02



Hermes guides

The tutor asks a question instead of giving the answer. The child fixes it themselves.

03



The company earns

Parent subscribes; a real Stripe test-mode charge clears on screen.

04



The company spends

Cut to the backend: the agent pays a mock server bill from a capped key — humans-on-loop panel visible.

Child-facing learning, then a hard cut to the backend autonomously earning and spending — the whole thesis in one shot.

Built for how this is scored.



Usefulness

A real, evidence-backed fix for a measured crisis — a child visibly learning, not marketing copy.



Viability

Near-zero operational overhead: the agent runs the business and buys its own capacity as it grows.



Presentation

A premium, on-brand product and a tight demo that lands the earn-and-spend loop in one cut.

Honest scope: the concept, research and design are complete; the hackathon build delivers one real end-to-end slice of the autonomous loop in Stripe test mode.

MØDE

Crafting mathematical minds with code adventures —
and a company that runs itself.



Evidence-led

Coding → math, taught
Socratically



Agent-run

Hermes earns, spends, operates



Safe by design

Bounded autonomy, child-data
compliant